EXAMPLE 7

Seven-Story, Plane Frame - Gravity and Lateral Loads Analysis

Problem Description

This is a seven-story plane frame. The frame is modeled with three column lines and two bays. Kip-inch-second units are used. Because the wide flange members used in the frame are older sections, their properties are not available in the AISC section property database included with the ETABS program, and the required properties therefore need to be explicitly provided in the input data.

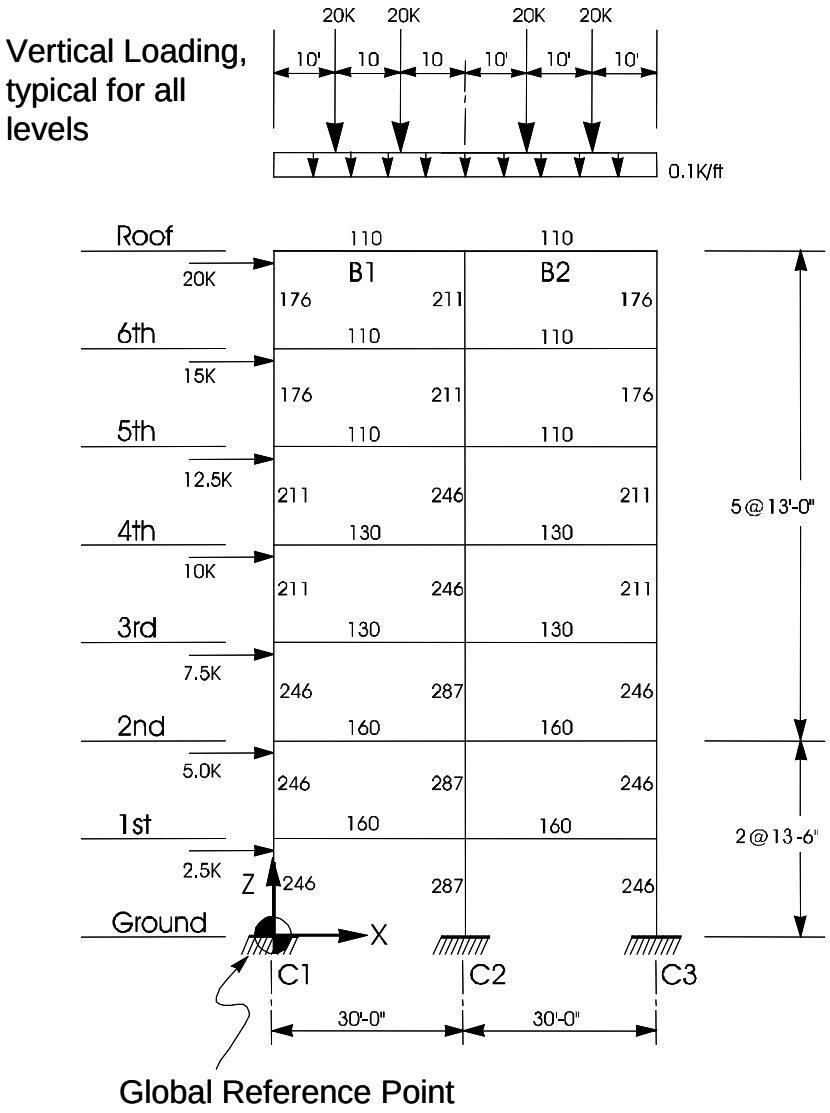
The example frame is analyzed in Wilson and Habibullah (1992) for gravity loads, static lateral loads and dynamic response spectrum loads. DYNAMIC/EASE2 analyzes the example frame under static lateral loads and dynamic response spectrum and time history loads. A comparison of key ETABS results with Wilson and Habibullah (1992) and DYNAMIC/EASE2 results is presented in Tables 7-1, 7-2, 7-3 and 7-4. Note the difference in modal combination techniques between ETABS and Wilson and Habibullah, which uses complete quadratic combination (CQC), and DYNAMIC/EASE2, which uses square root of the sum of the squares combination (SRSS).

Geometry, Properties and Loads

The gravity loads and the geometry of the frame are shown in Figure 7-1.

The frame is subjected to the following lateral loads:

- Static lateral loads, shown in Figure 7-1
- Lateral loads resulting from the El Centro 1940 N-S component seismic response spectra, 5 percent damping
- Lateral loads resulting from the El Centro 1940 N-S component acceleration time history



All columns are W14s
All beams are W24s
Member weights are indicated
Typical story mass = 0.49 kip-sec^2/in

Figure 7-1 Seven-Story Plane Frame

Technical Features of ETABS Tested

- Two-dimensional frame analysis
- User-specified section properties
- User-specified lateral loads
- Dynamic response spectrum analysis
- Dynamic time history analysis

Results Comparison

The comparison of the results for all three analyses is excellent.

Table 7-1 Comparison of Results for Static Lateral Loads

Quantity	ETABS	Wilson and Habibullah	DYNAMIC/EASE2
Lateral Displacement at Roof	1.4508	1.4508	1.4508
Axial Force Column C1, at ground	69.99	69.99	69.99
Moment Column C1, at ground	2324.68	2324.68	2324.68

Table 7-2 Comparison of Results for Periods of Vibration

Mode	ETABS	Wilson and Habibullah	DYNAMIC/EASE2
1	1.27321	1.27321	1.27321
2	0.43128	0.43128	0.43128
3	0.24205	0.24204	0.24204
4	0.16018	0.16018	0.16018
5	0.11899	0.11899	0.11899
6	0.09506	0.09506	0.09506
7	0.07952	0.07951	0.07951

Table 7-3 Comparison of Results for Response Spectrum Analysis

Quantity	ETABS CQC Combination	Wilson and Habibullah CQC Combination	DYNAMIC/EASE2 SRSS Combination
Lateral Displacement at Roof	5.4314	5.4314	5.4378
Axial Force Column C1 at ground	261.52	261.50	261.76
Moment Column C1 at ground	9916.12	9916.11	9868.25

Table 7-4 Comparison of Results for Time History Analysis

Quantity	ETABS	Wilson and Habibullah
Maximum Roof Displacement	5.49	5.48
Maximum Base Shear	285	284
Maximum Axial Force, Column C1 at ground	263	258
Maximum Moment, Column C1 at ground	9104	8740

Computer Files

The input data file is Example 07.EDB. The input history is ELCN-THU. Time history results are obtained for the first eight seconds of the excitation. This is consistent with DYNAMIC/EASE2, with which the results are compared. These computer files are provided as part of the ETABS installation.

Conclusions

Noting the difference in modal combination techniques between ETABS and Wilson and Habibullah, which uses complete quadratic combination (CQC), and DYNAMIC/EASE2, which uses square root of the sum of the squares combination (SRSS), the results of the testing are acceptable.